

Qasim Hussain

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SKILLS

Wet Lab

Cell Culture (cell freezing, cell thawing, cell passaging, cell counting and cloning) | Virus Culture | *In Vitro* Assays (cell viability assay, LDH assay, virus inhibition assay, G/F actin assay) | RNA and DNA Extraction | RT-qPCR | Gel Electrophoresis | Bacterial Culture | Protein production and purification | SDS PAGE | Western Blot | Confocal Microscopy | Serum biochemistry (AST, ALT, triglycerides, total cholesterol and glucose) | Cryosectioning of tissue samples | Handling, restraint, and physiological maintenance of murine (mice) and avian (pigeon) experimental models | Compliance with animal welfare protocols and execution of controlled dosing regimens, etc.

Dry Lab (NGS Analysis)

Whole Genome Sequencing (Short Reads, Long Reads, Hybrid) | Whole Exome Sequencing | Bulk RNA-Seq | Single Cell RNA-Seq | Spatial Transcriptomics | Pangenomics | Antibiotic Resistance Genes Detection (AMR) Analysis | CAZyme Analysis | Secretion System Detection | CRISPR-Cas | COJScan and TFFscan System Analysis | Biosynthetic Gene Cluster Analysis | ImmunoPhagomics | WGCNA | PanGWAS | TWAS | Ribo-Seq | Amplicon Sequencing (16S, 18S and ITS) | Metagenomics, etc.

Dry Lab (Structural Bioinformatics)

Molecular Docking | Molecular Dynamics Simulation | Virtual Screening | Protein Structure Prediction and Modeling | *De Novo* Protein Design | *De Novo* Binder Prediction | *De Novo* Aptamers Prediction, etc.

Data Analysis, Statistics & Machine Learning

Statistical Analysis (R, Python, GraphPad Prism) | Hypothesis Testing (Parametric/Non-parametric) | Data Visualisation | Machine Learning

Programming & Workflow Management

Markdown | Python | R | Linux/Bash | SQL | Git | Conda | Snakemake, etc.

EDUCATION & TRAINING

[9 Sep 2024 - 10 Jul 2026] **MS Animal Vaccine Technology**

National Pingtung University of Science and Technology

Final grade: 93.63% | **Thesis:** *In silico* drug discovery via high-throughput virtual screening and *in vitro* validation of novel antivirals of Porcine Epidemic Diarrhea Virus (PEDV).

[4 Sep 2019 - 31 Aug 2023] **BS Zoology**

University of Education, Township, Lahore

WORK EXPERIENCE

[13 Aug 2026 - Current] **Research Assistant (Full-time) | Cardiovascular Research Group**

Mackay Memorial Hospital

[9 Sep 2024 - 10 Jul 2026] **Graduate Researcher | Graduate Institute of Animal Vaccine Technology**

National Pingtung University of Science and Technology

[21 Dec 2025 - Current] **Independent Computational Research**

Portfolio

PUBLICATIONS

1. **Lee, C.M., Villanueva, B.H.A., Minh, H., Hussain, Q.*, & Chuang, K.-P.* (2025). Evaluation of feline exosome mediated renal regeneration in adenine-induced chronic kidney disease. [co-corresponding author]**
2. **Villanueva, B.H.A., Gomez, M.C., Hussain, Q., Hong, Q.D., & Chuang, K.-P. (2025). Conserved internal PB2–PB1–PA–NP–M genes and divergent HA/NA in bat influenza A viruses H9N2, H17N10, and H18N11.**
3. **Villanueva, B.H.A., Hussain, Q., Hong, Q.D., Chen, M.H., Minh, H., Huang, H.Y., Tyan, Y.C., & Chuang, K.-P. Emergence of Black Queen Cell Virus in *Apis mellifera ligustica* of southern Taiwan: evidence from local genomic surveillance. (Under Review)**
4. **Nadeem, S., Tahir, S.B., Ahmad, M., Aslam, A., Farooq, M., Hussain, Q., Majid, H., Usman, S.M., Bashir, S., & Maqsood, H. (2023). The next pandemic: a comprehensive look into zoonotic emerging threats. (Book Chapter)**
5. **Afzal, H., Umar, M., Hussain, Q., & Cheng, L.-T.* Evaluating the adjuvanticity of N-D1 domain of Flagellin encompassing hot-spot region in combination with Porcine Epidemic Diarrhea Virus (PEDV). (Under Review)**
6. **Are *de novo* designed proteins immunogenic? (Review: In Preparation)**

CONFERENCES & SEMINARS

1. **Umar, M., Hussain, Q., & Cheng, L.-T.* "From coexistence to contagion: assessing zoonotic threats from Formosan Macaques in urban Taiwan." 2025 Grand Challenge Taiwan SDGs Issue English Poster Competition, NCKU. (Poster, 2nd author)**
2. **Hussain, Q., & Chuang, K.-P.* "*In vitro* antiviral efficacy of Molnupiravir against Porcine Epidemic Diarrhea Virus (PEDV)." 5th International Symposium on Immunotechnology and Animal Biologics Development, NPUST. (Poster, 1st author)**
3. **Duong, H.Q., Hussain, Q., Nguyen, D.L., & Chuang, K.-P.* "Potential application of plant-derived nanoparticles as antiviral agents against Parvovirus." 2026 International Nanotechnology Olympiad, Academia Sinica. (Poster, 2nd author)**
4. **Hussain, Q., Villanueva, B.H.A., Chuang, K.-P.*, & Hsin, K.T.* "*In silico* drug discovery via high-throughput virtual screening and *in vitro* validation of novel antivirals against Porcine Epidemic Diarrhea Virus (PEDV) Spike Protein." 2026 International Symposium on Sustainable Innovation in Smart Agriculture (ISSI-SA 2026), NPUST. (Poster, 1st author)**

HONOURS & AWARDS

[13 Jun 2026] **Outstanding Researcher Award** (**Awarding institution:** National Pingtung University of Science and Technology)

[17 Jan 2026] **Chung Hwa Rotary Educational Foundation Scholarship Holder** (**Awarding institution:** Chung Hwa Rotary Educational Foundation)

[4 Sep 2019] **University Scholarship Holder** (**Awarding institution:** University of Education, Township, Lahore)

[11 Sep 2017] **Prime Minister Laptop** (**Awarding institution:** Punjab Board of Technical Education)

[21 Jul 2017] **College Scholarship Holder** (**Awarding institution:** Webster Group of Colleges)

[11 May 2015] **Merit Scholarship Holder** (**Awarding institution:** Punjab Educational Endowment Fund (PEEF))

LANGUAGE SKILLS

English (IELTS: In Preparation)